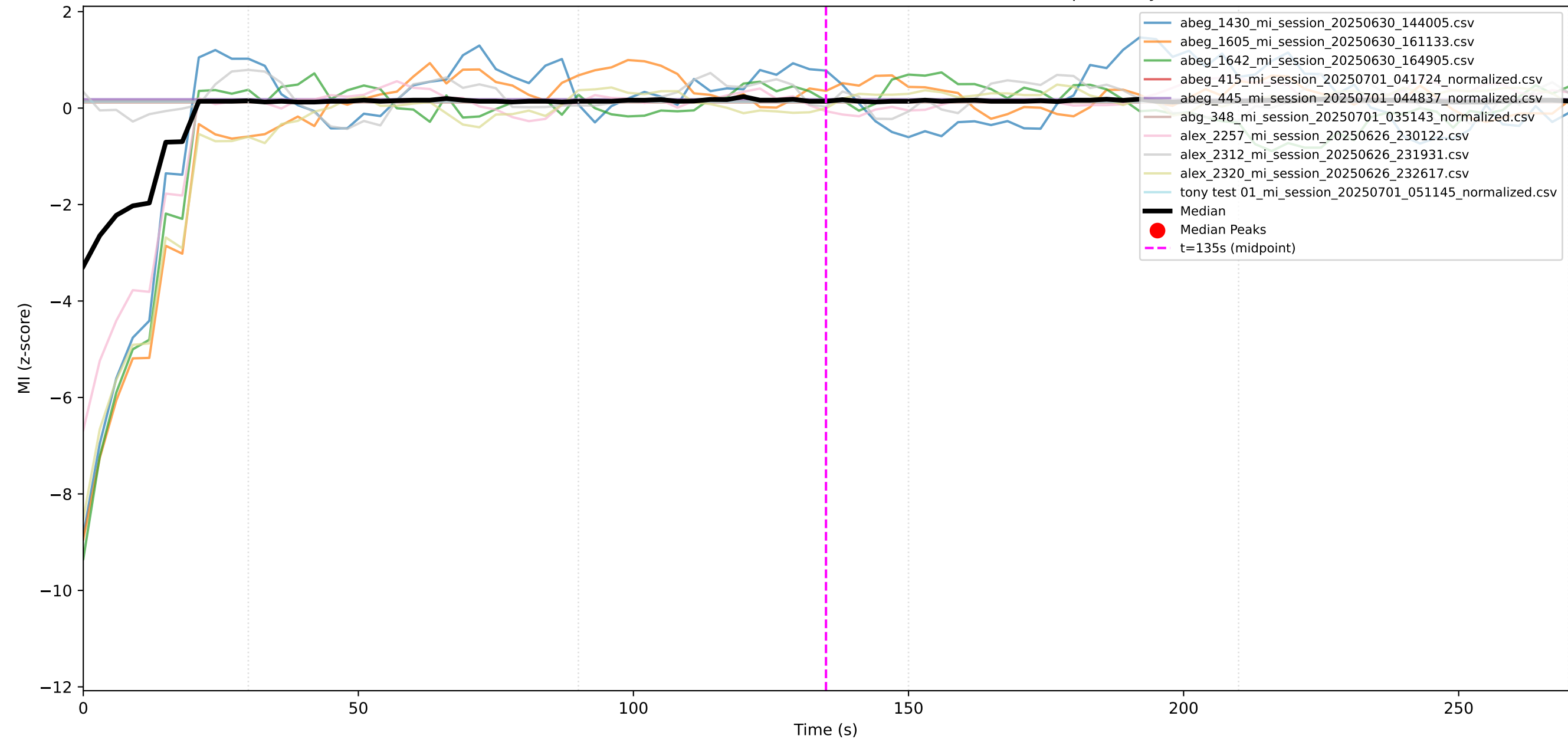
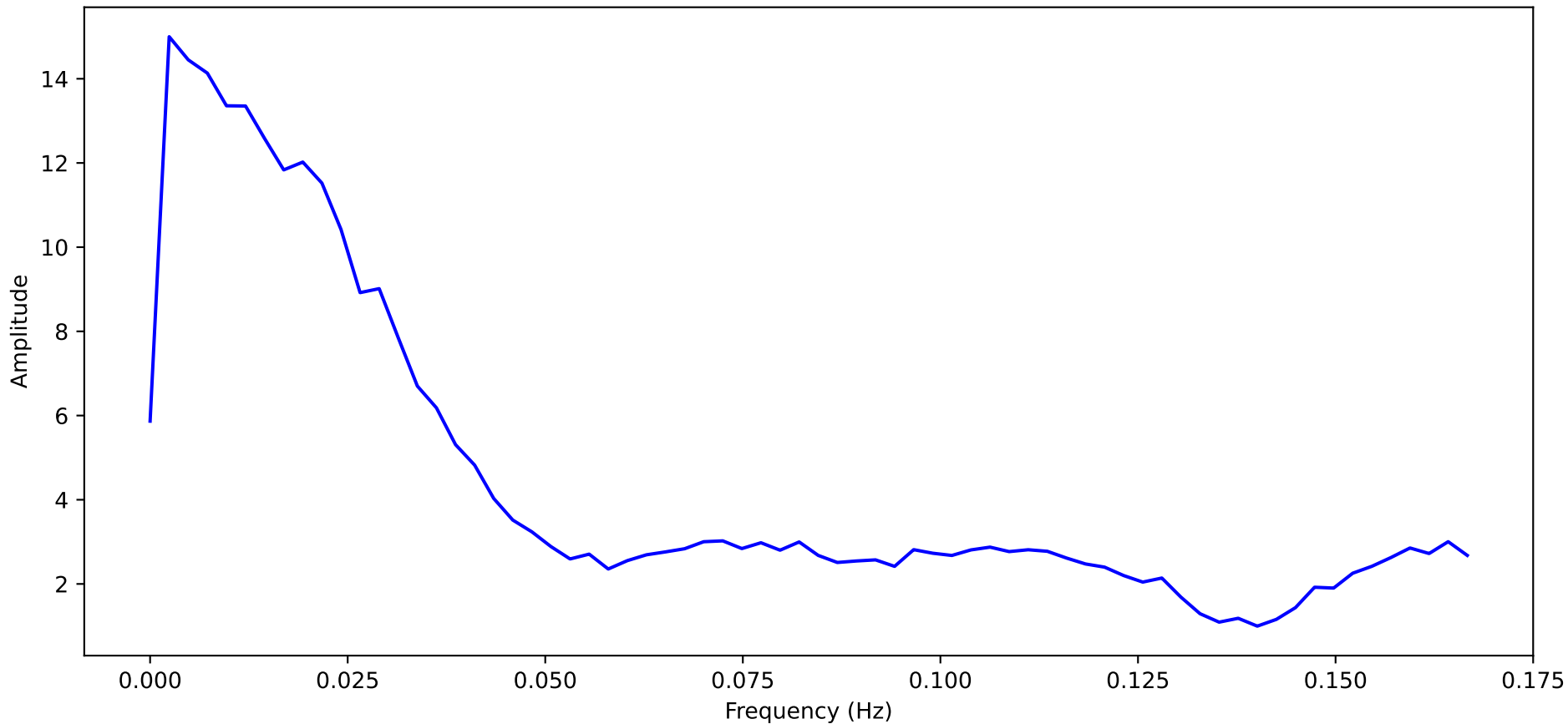


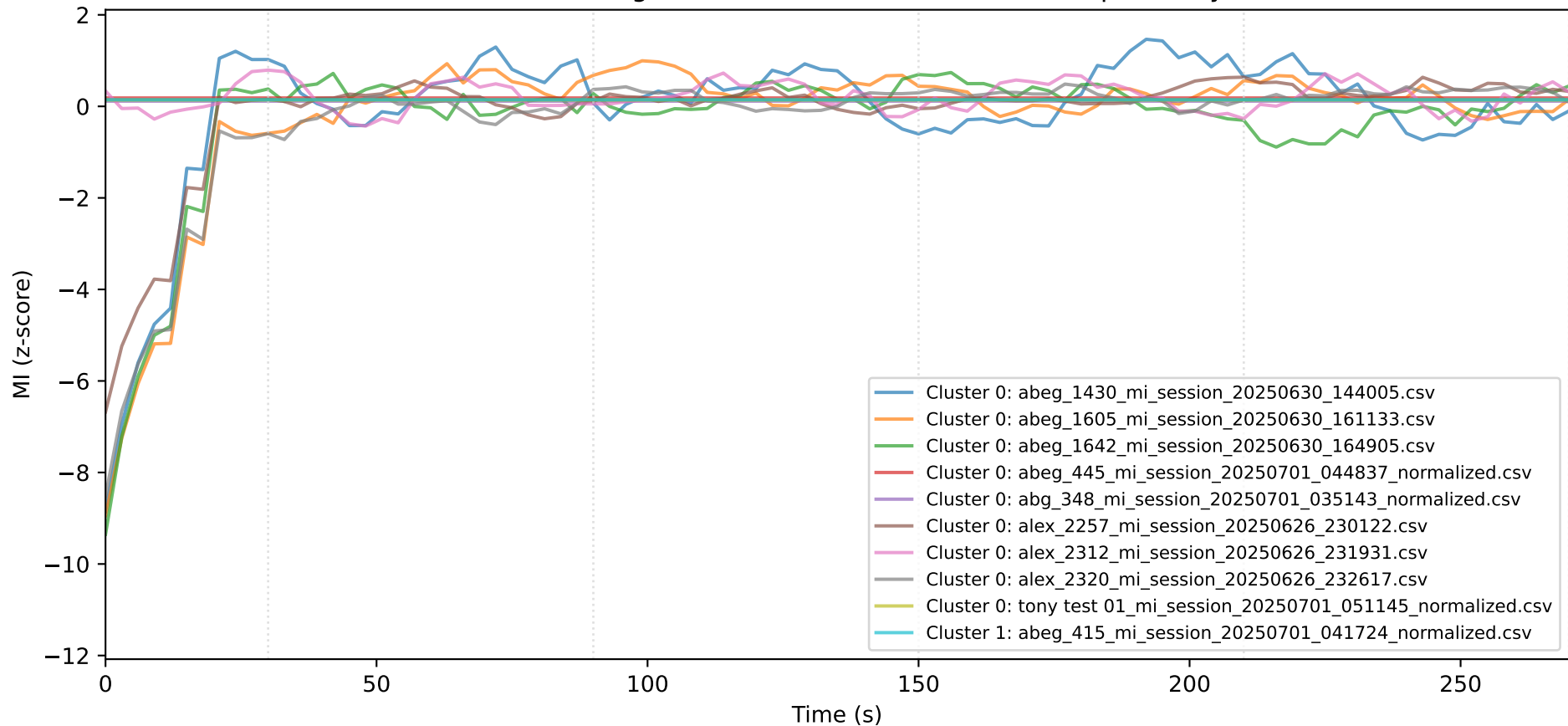
MI Curves (z-score, MA=20) with Median and Peaks (270s, 1 sample every 3s)



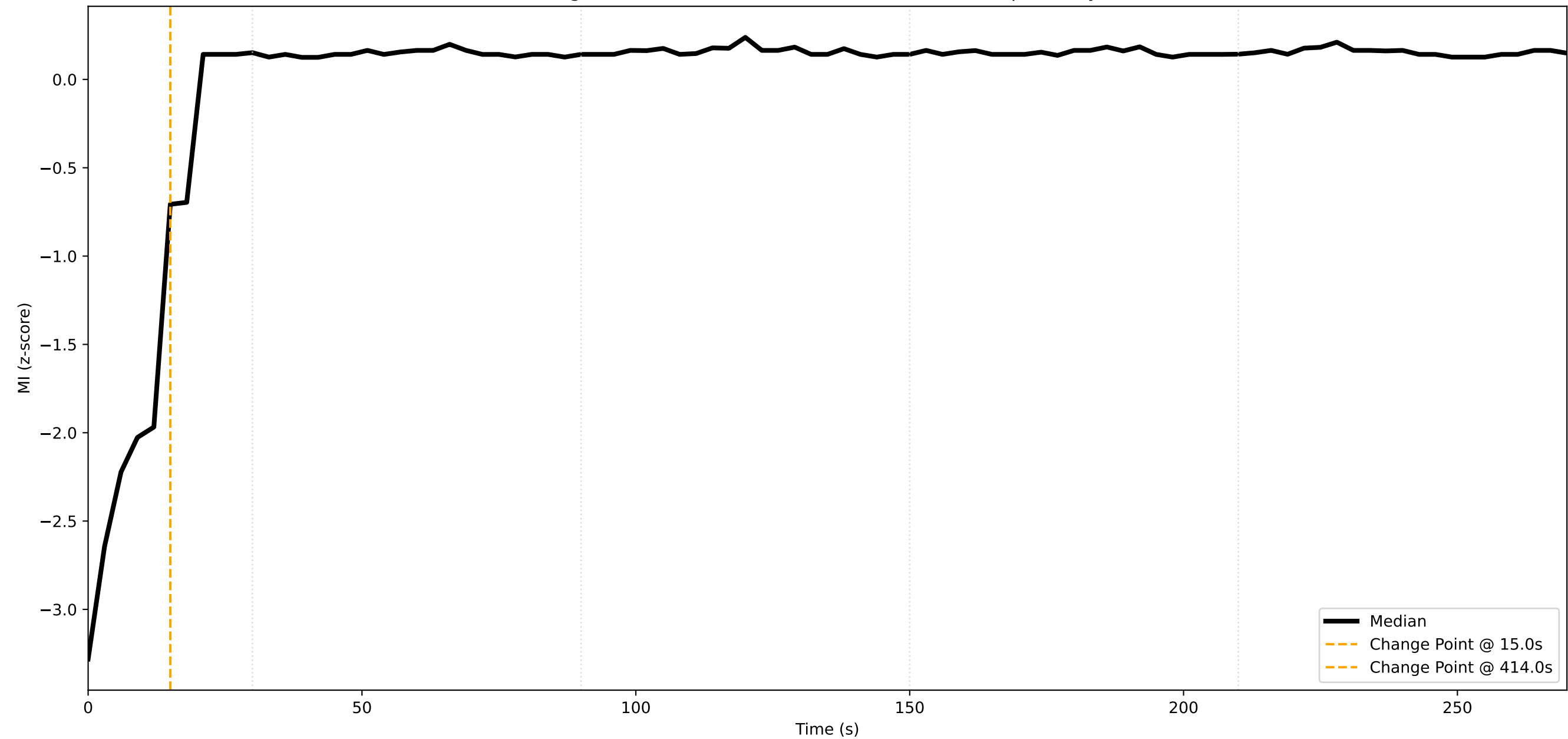
Median MI Frequency Spectrum (FFT)



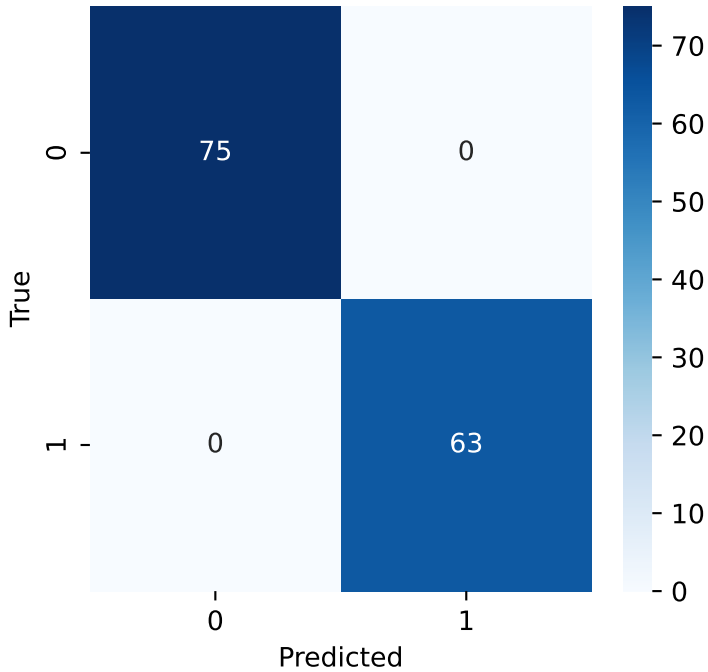
KMeans Clustering of MI Curves (k=2) - 270s (1 sample every 3s)



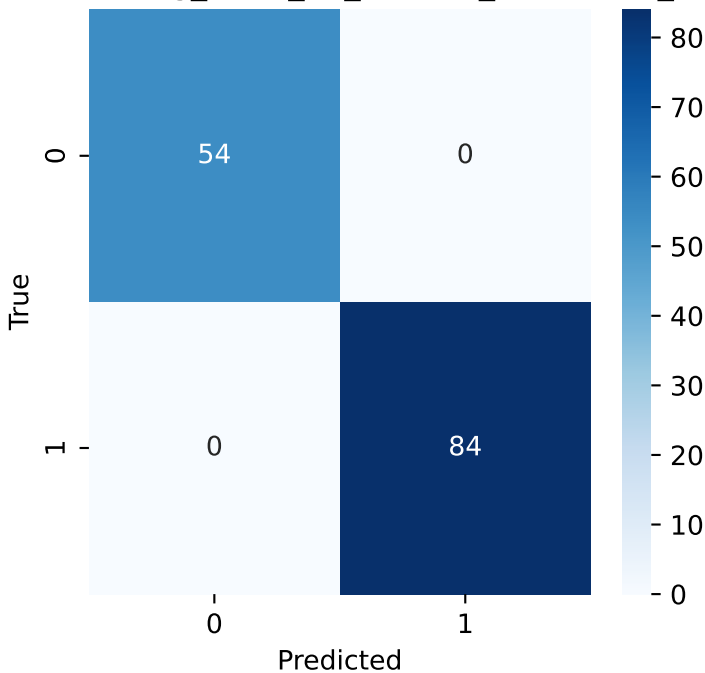
Change Point Detection on Median MI - 270s (1 sample every 3s)



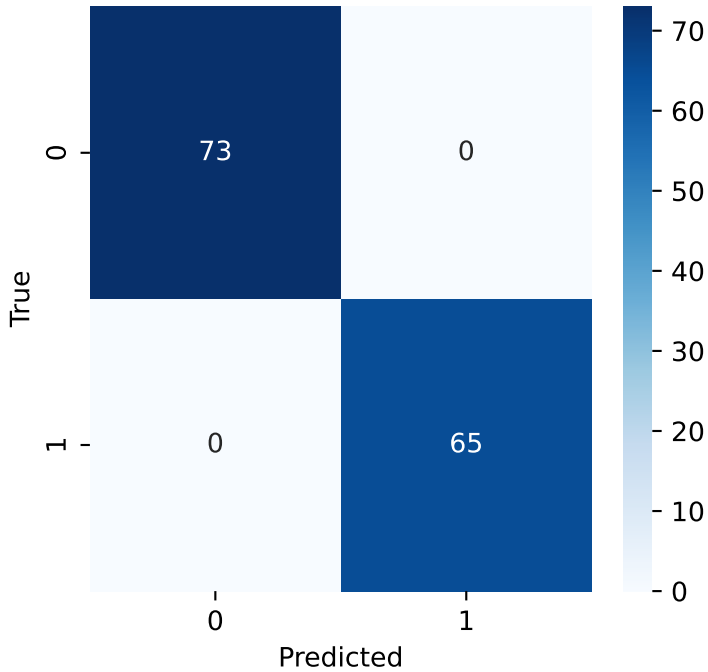
Matrix: abeg_1430_mi_session_20250630_1440



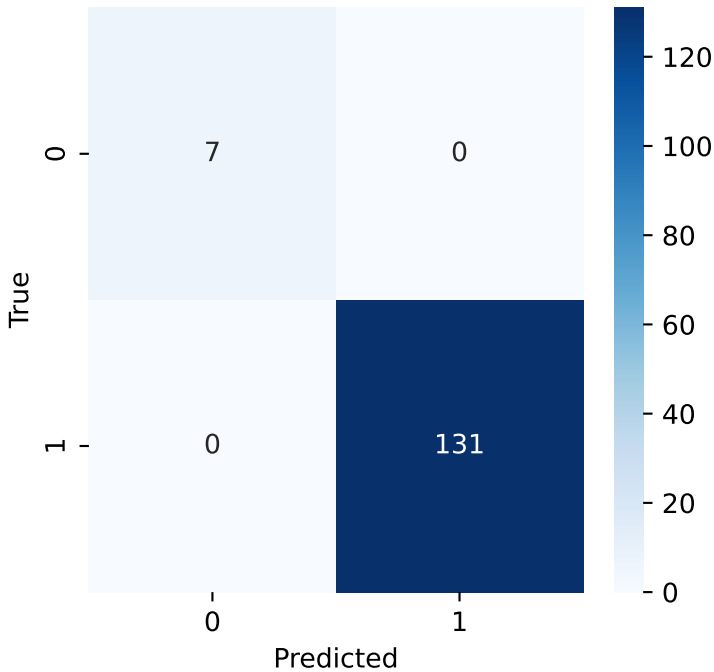
Matrix: abeg_1605_mi_session_20250630_1611



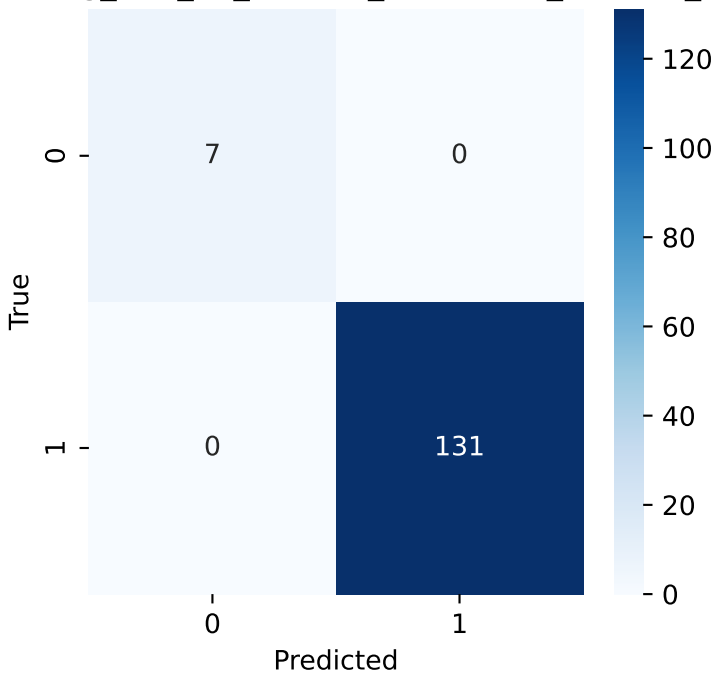
Matrix: abeg_1642_mi_session_20250630_1649



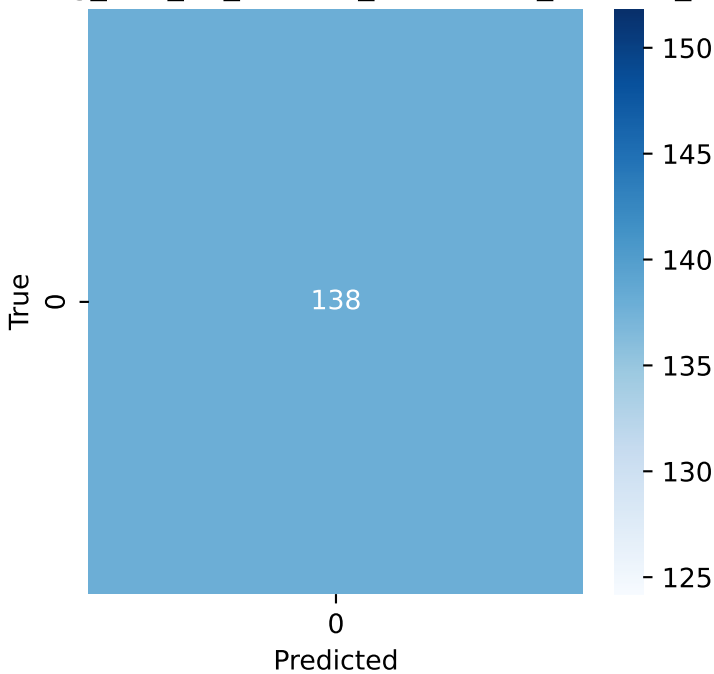
x: abeg_415_mi_session_20250701_041724_no



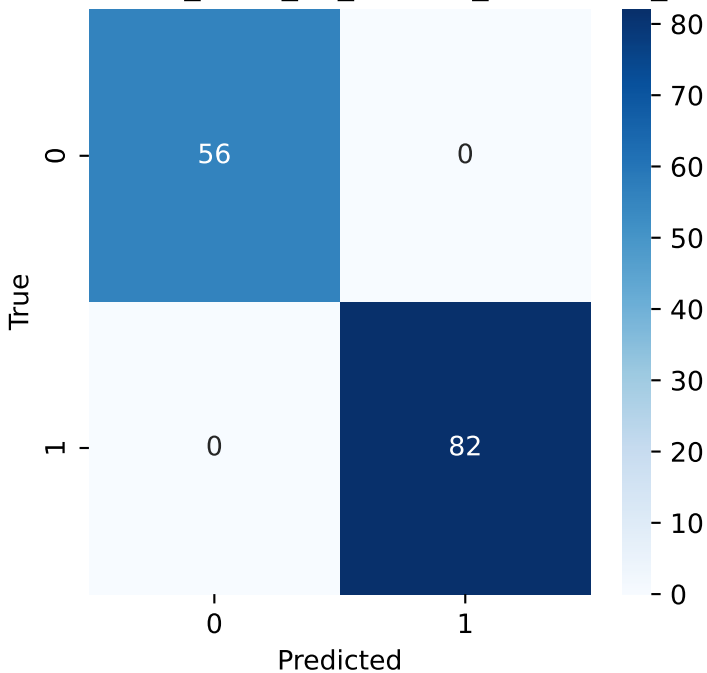
x: abeg_445_mi_session_20250701_044837_no



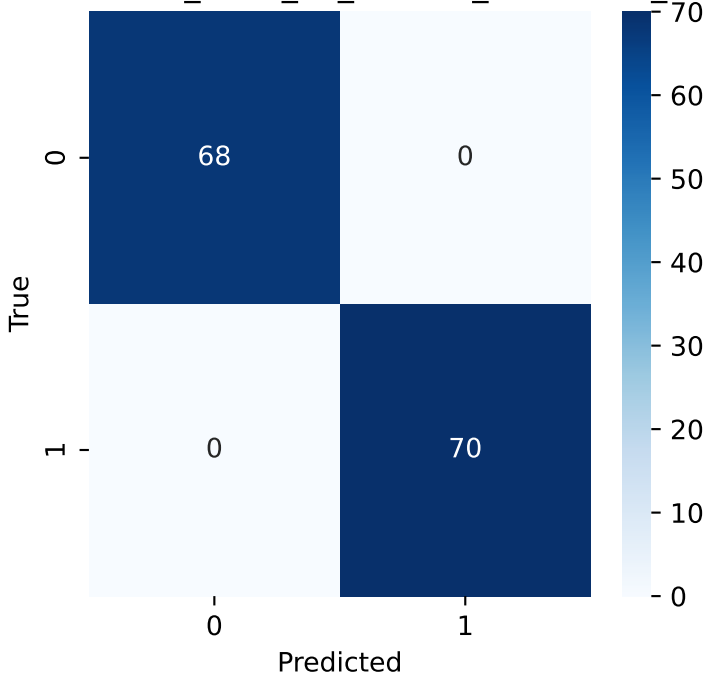
ix: abg_348_mi_session_20250701_035143_nor



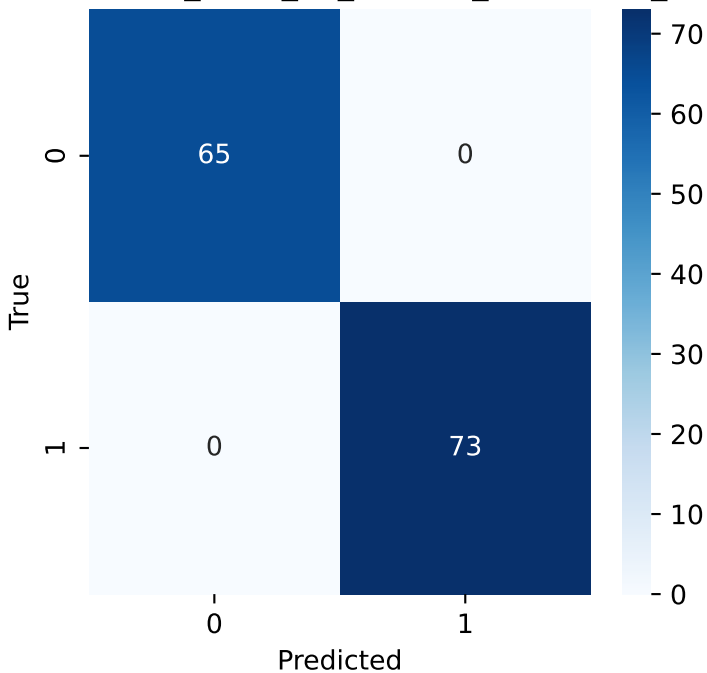
h Matrix: alex_2257_mi_session_20250626_2301



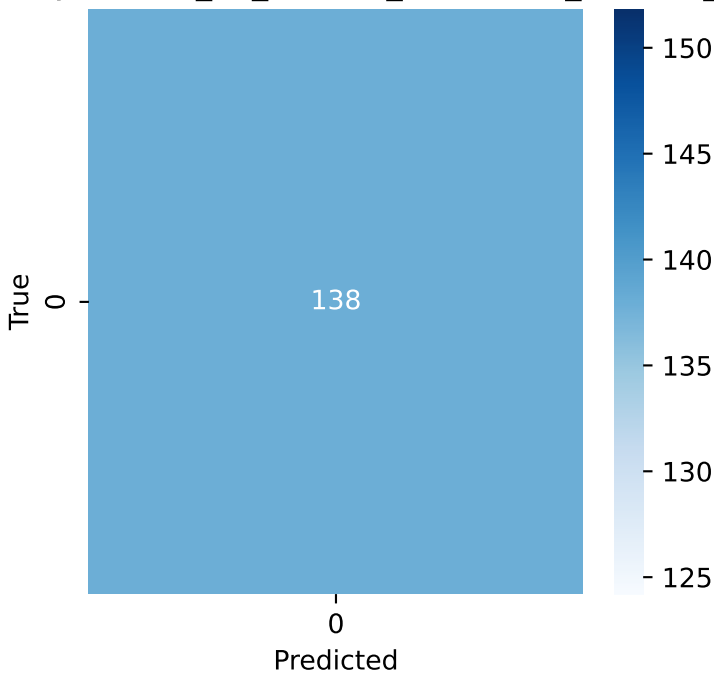
h Matrix: alex_2312_mi_session_20250626_2319



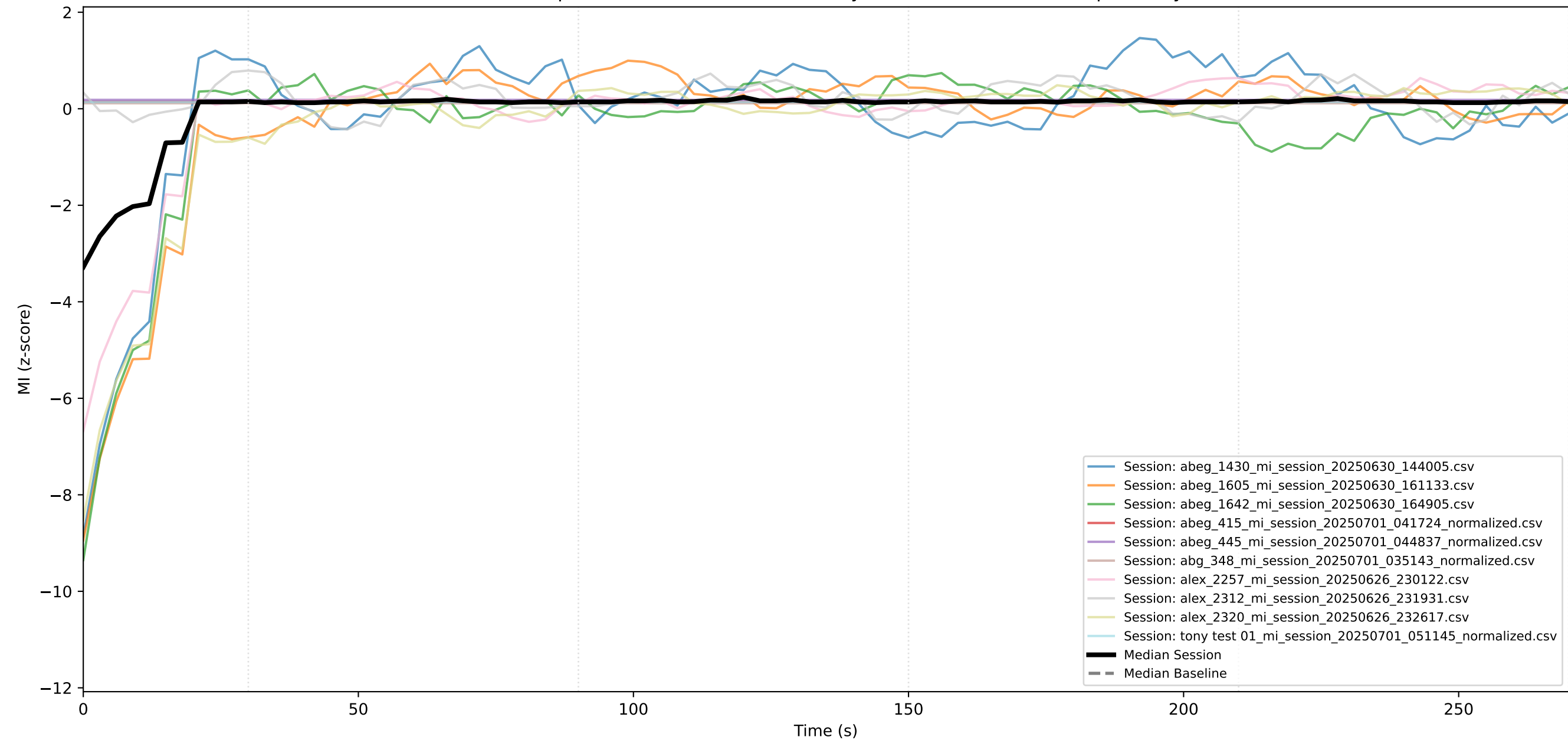
h Matrix: alex_2320_mi_session_20250626_2326



: tony test 01_mi_session_20250701_051145_no



Session MI (up to 10) vs. Median Baseline (Eyes Closed) - 270s (1 sample every 3s)



MI Advanced Report

Generated: 20250717_025651

MI Advanced Report - Statistical Summary (1 sample every 3 seconds)

1. Mindfulness Index (MI) Trajectory

- * All MI session files with ≥ 34 samples were included (100s minimum at 1 sample per 3s).
- * Individual and median MI curves were plotted with 20-second moving average (see Figure 8.1).
- * RM-ANOVA across six epochs (0-45s, 45-90s, 90-135s, 135-180s, 180-225s, 225-270s):
Anova

```
=====
      F Value Num DF  Den DF Pr > F
-----
Epoch  7.7459 5.0000 45.0000 0.0000
=====
```

- * Post-hoc pairwise comparisons (Tukey HSD):
Multiple Comparison of Means - Tukey HSD, FWER=0.05

```
=====
group1  group2 meandiff p-adj  lower  upper  reject
-----
  0-45s 135-180s   1.2073 0.0002   0.4609 1.9537   True
  0-45s 180-225s   1.2854 0.0001    0.539 2.0318   True
  0-45s 225-270s   1.1805 0.0003   0.4341 1.9269   True
  0-45s  45-90s   1.2369 0.0001   0.4905 1.9833   True
  0-45s  90-135s   1.2831 0.0001   0.5367 2.0295   True
135-180s 180-225s   0.0781 0.9996  -0.6684 0.8245  False
135-180s 225-270s  -0.0268    1.0  -0.7732 0.7196  False
135-180s  45-90s   0.0296    1.0  -0.7168 0.776   False
135-180s  90-135s   0.0758 0.9997  -0.6706 0.8222  False
180-225s 225-270s  -0.1048 0.9983  -0.8512 0.6416  False
180-225s  45-90s  -0.0484    1.0  -0.7948 0.698   False
180-225s  90-135s  -0.0023    1.0  -0.7487 0.7441  False
225-270s  45-90s   0.0564 0.9999   -0.69 0.8028  False
225-270s  90-135s   0.1025 0.9985  -0.6439 0.8489  False
  45-90s  90-135s   0.0461    1.0  -0.7003 0.7926  False
=====
```

- * Time-frequency analysis (FFT) of the median MI curve was performed (see Figure 8.3).

2. MI Dynamics

- * Peak analysis (median curve): 0 peaks ($z > 1.5$) at sample times [], amplitudes [], durations [] (see Table 8.2, Figure 8.4).
- * AUC (median): 7.43; mean $\pm$ SD (individual): 2.66 $\pm$ 16.46 (see Table 8.3).
- * Bootstrap 95% CI for mean MI difference (0-135s vs 136-270s): mean diff = 0.000, CI = [0.000, 0.000].
- * Change point detection (Pelt algorithm): 1 change points at sample indices [5] (see Figure 8.5).

3. Clustering & Mixed-Effects

- * K-Means clustering (k=2): cluster averages {}, user distribution {0: 9, 1: 1} (see Figure 8.6).

* Mixed-effects model (MI ~ time + (1|Subject)): Mixed Linear Model Regression Results

```
=====
Model:          MixedLM Dependent Variable: value
No. Observations: 1380   Method:          REML
No. Groups:      10      Scale:           1.1131
Min. group size: 138     Log-Likelihood: -2044.2032
Max. group size: 138     Converged:      Yes
Mean group size: 138.0
=====
      Coef.  Std.Err.   z    P>|z| [0.025 0.975]
-----
Intercept  -0.249    0.066  -3.755 0.000  -0.380 -0.119
time       0.004    0.001   5.123 0.000   0.002  0.005
file Var   0.012    0.009
=====
```